

POWER BI

Smart Narrative & DAX Measures

Technical Documentation

Dataset: AdventureWorksDW2022

Tool: Power BI Desktop / Service

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1. Overview

This document covers the design, logic, and implementation of the **Smart Narrative DAX measure framework** built for Power BI on the AdventureWorksDW2022 data model. The framework delivers AI-style executive narratives that update dynamically with slicer context, surfacing financial performance, regional intelligence, product insights, and strategic risk signals in natural language — with zero manual editing required.

Narrative Measures Covered

Measure	Purpose	Section
Executive Narrative	Concise per-KPI executive summary with MoM context	3
Regional Context Narrative	Regional variance, YoY growth and consistency analysis	4
Executive Summary	Full multi-section strategic intelligence brief	5

Supporting Concept Layers

Concept	Key Measures	Section
Strategic Health Score	Composite 0–10 score across 5 dimensions	6
Regional Growth Consistency	Consistency rate, budget adherence, avg YoY per region	7
BenchmarkTable	Centralized industry benchmark lookup table	8

2. Data Model

2.1 Core Tables

Table	Role	Key Columns Used
FactInternetSales	Sales transactions	SalesAmount, OrderQuantity, UnitPrice, ProductStandardCost, OrderDateKey, SalesTerritoryKey, CustomerKey
FactInternetSalesTarget	Budget targets	SalesTarget — granularity: Country, Category, BudgetMonth
DimDate	Time intelligence	Date, CalendarYear, MonthNumberOfYear — must be marked as Date Table
DimSalesTerritory	Geographic hierarchy	SalesTerritoryKey, SalesTerritoryCountry, Group
DimProduct	Product master	ProductKey, ProductSubcategoryKey, Subcategory, ProductCategoryKey, Category, ListPrice
DimCustomer	Customer master	CustomerKey
BenchmarkTable	Industry KPI benchmarks	Metric, Benchmark — see Section 8

2.2 Required Relationships

From	To (Primary Key)
FactInternetSales[OrderDateKey]	DimDate[DateKey]
FactInternetSales[SalesTerritoryKey]	DimSalesTerritory[SalesTerritoryKey]
FactInternetSales[ProductKey]	DimProduct[ProductKey]
FactInternetSales[CustomerKey]	DimCustomer[CustomerKey]
FactInternetSalesTarget → DimDate, DimSalesTerritory, DimProduct	Via respective keys at country/category/month granularity

NOTE

DimDate must be marked as a Date Table in Power BI (Table Tools → Mark as Date Table) with DimDate[Date] as the date column. This is mandatory for SAMEPERIODLASTYEAR and DATEADD to function correctly.

3. Executive Narrative

3.1 Purpose

The **Executive Narrative** measure produces a concise, four-line card-ready summary covering revenue performance, order momentum, gross margin health, and a dynamic risk alert. It is designed for a **Smart Narrative or Card visual** at the top of any executive dashboard page, updating fully with Year, Month, and Country slicer selections.

3.2 Dependency Measures

Measure	Formula Summary	Used In Narrative
Total Revenue	SUM(FactInternetSales[Sales Amount])	Revenue figure
Total Revenue PM	CALCULATE([Total Revenue], DATEADD(DimDate[Date],-1,MONTH))	MoM base
Revenue MoM %	DIVIDE([Total Revenue]-[Total Revenue PM],[Total Revenue PM])	MoM % line
Total Orders	DISTINCTCOUNT(FactInternetSales[SalesOrderNumber])	Order count
Orders MoM %	DIVIDE([Total Orders]-[Total Orders PM],[Total Orders PM])	Order direction
Gross Profit	SUM(GrossProfit)	GP figure
Gross Profit Margin %	DIVIDE([Gross Profit],[Total Revenue])	Margin line
Gross Profit Margin % PM	CALCULATE([Gross Profit Margin %], DATEADD(-1,MONTH))	Margin base
Margin Change PP	[Gross Profit Margin %] - [Gross Profit Margin % PM]	PP shift
Selected Country	HASONEVALUE check on DimSalesTerritory[Country]	Scope label
Selected Year	HASONEVALUE check on DimDate[Year]	Scope label
Selected Month	HASONEVALUE check on DimDate[MonthNo]	Scope label

3.3 Narrative Measure — Full DAX

```
Measure 'Executive Narrative' =  
VAR _Rev = [Total Revenue]  
VAR _RevMoM = [Revenue MoM %]  
VAR _Orders = [Total Orders]  
VAR _OrdersMoM = [Orders MoM %]  
VAR _GPM = [Gross Profit Margin %]  
VAR _MarginPP = [Margin Change PP]  
VAR _Country = [Selected Country]  
VAR _Year = [Selected Year]  
VAR _Month = [Selected Month]
```

```

VAR _Scope = "" & _Country & " | " & _Year & " | Month " & _Month
VAR _Perf = "☐ Revenue " & IF(_RevMoM>=0,"▲ ","▼ ")
           & FORMAT(ABS(_RevMoM),"0.0%") & " MoM to "
           & FORMAT(_Rev,"$#,0.0,") & "M. Orders "
           & IF(_OrdersMoM>=0,"▲ ","▼ ") & FORMAT(ABS(_OrdersMoM),"0.0%") & "."
VAR _Margin = "☐ Gross margin: " & FORMAT(_GPM,"0.0%")
              & " (" & IF(_MarginPP>=0,"▲ ","▼ ")
              & FORMAT(ABS(_MarginPP),"0.0%") & " pp vs prior month)."
VAR _Alert = "☐ " & SWITCH(TRUE(),
    _MarginPP < -0.02, "Margin compression >2pp – strategic review required.",
    _MarginPP < 0,     "Margin softening – monitor closely.",
    _RevMoM < 0,       "Revenue declining – assess demand signals.",
    "No critical concerns. Performance on track.")
RETURN
IF(ISBLANK(_Rev), "☐ No data for selected filters.",
    _Scope & UNICHAR(10) & _Perf & UNICHAR(10) & _Margin & UNICHAR(10) & _Alert)

```

3.4 Output Structure

Line	Content
☐ Scope	Active country / year / month filter context
☐ Performance	Revenue MoM direction + figure, Order MoM direction
☐ Margin	Current gross margin % + pp change vs prior month
☐ Alert	Dynamic risk signal — escalates based on severity thresholds

4. Regional Context Narrative

4.1 Purpose

The **Regional Context Narrative** extends the executive narrative to surface geographic intelligence — identifying the top variance carrier, highest-margin region, fastest/slowest YoY growers, and a context-aware drill-in for the currently selected region. It also incorporates the Regional Growth Consistency Index (see Section 7).

4.2 Additional Dependency Measures

Measure	Formula Summary	Purpose
Total Budget	SUM(FactInternetSalesTarget[SalesTarget])	Budget base
Revenue Variance	[Total Revenue] - [Budget Revenue]	Variance \$
Revenue Variance %	DIVIDE([Budget Variance],[Budget Revenue])	Variance %
Total Variance All Regions	CALCULATE(ABS([Budget Variance]), ALL(DimSalesTerritory))	Denominator for share
Region Variance Share %	DIVIDE(ABS([Budget Variance]),[Total Variance All Regions])	Region share of total variance
Total Revenue All Regions	CALCULATE([Total Revenue], ALL(DimSalesTerritory))	Revenue denominator
Region Revenue Share %	DIVIDE([Total Revenue],[Total Revenue All Regions])	Region revenue share
Total Revenue PY	CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(DimDate[Date]))	YoY base
Revenue YoY %	DIVIDE([Total Revenue]-[Total Revenue PY],[Total Revenue PY])	YoY growth rate
Top Region by YoY Growth	FIRSTNONBLANK via TOPN on Revenue YoY %	Best YoY region name
Weakest Region by Budget Variance	FIRSTNONBLANK via TOPN ASC on Budget Variance	Worst variance region
Top Region by Margin	FIRSTNONBLANK via TOPN on Gross Profit Margin %	Highest margin region
Most Consistent Region	TOPN on Regional Growth Consistency Score	Most reliable region
Least Consistent Region	TOPN ASC on Regional Growth Consistency Score	Most volatile region
Avg Regional Consistency Score	AVERAGEX across Group	Portfolio average score

4.3 Narrative Measure — Full DAX

```

Measure 'Regional Context Narrative' =
VAR _TopVarRegion      = [Top Region by Variance Share]
VAR _TopVarPct         = [Top Region Variance Share %]
VAR _TopMarginRegion   = [Top Region by Margin]
VAR _TopMargin         = [Top Region Margin]
VAR _RevShare          = [Region Revenue Share %]
VAR _Variance          = [Budget Variance]
VAR _VariancePct       = [Budget Variance %]
VAR _YoY               = [Revenue YoY %]
VAR _TopYoYRegion      = [Top Region by YoY Growth]
VAR _TopYoYPct         = [Top Region YoY %]
VAR _WeakYoYRegion     = [Weakest Region by YoY Growth]
VAR _WeakYoYPct        = [Weakest Region YoY %]
VAR _MostConsistent    = [Most Consistent Region]
VAR _MostScore          = [Most Consistent Region Score]
VAR _MostPct           = [Most Consistent Region Consistency %]
VAR _LeastConsistent   = [Least Consistent Region]
VAR _LeastScore         = [Least Consistent Region Score]
VAR _LeastBudget        = [Least Consistent Region Budget Adherence]
VAR _AvgConsistency     = [Avg Regional Consistency Score]
VAR _SelectedRegion     = IF(HASONEVALUE(DimSalesTerritory[Group]),
                             VALUES(DimSalesTerritory[Group]),
                             "All Regions")

VAR _Line1 = "" & _TopVarRegion & " carries "
              & FORMAT(_TopVarPct,"0%") & " of total variance - "
              & IF(_Variance>=0,"▲ surplus ", "▼ shortfall ")
              & FORMAT(ABS(_Variance),"$#,0.0,,") & "M ("
              & FORMAT(ABS(_VariancePct),"0.0%") & " vs budget). "
VAR _Line2 = "" & _TopMarginRegion & " leads on margin at "
              & FORMAT(_TopMargin,"0.0%") & ". Overall revenue "
              & IF(_YoY>=0,"▲ ", "▼ ") & FORMAT(ABS(_YoY),"0.0%") & " YoY."
VAR _Line3 = "☐ Fastest: " & _TopYoYRegion
              & " (" & IF(_TopYoYPct>=0,"+", "") & FORMAT(_TopYoYPct,"0.0%") & " YoY). "
              & "Lagging: " & _WeakYoYRegion
              & " (" & IF(_WeakYoYPct>=0,"+", "") & FORMAT(_WeakYoYPct,"0.0%") & " YoY). "
VAR _Line4 = "☐ Consistency Index (avg " & FORMAT(_AvgConsistency,"0.0") & "/10): "
              & _MostConsistent & " most reliable (" & FORMAT(_MostScore,"0.0")
              & "/10, " & FORMAT(_MostPct,"0%") & " months positive YoY). "
              & _LeastConsistent & " most volatile (" & FORMAT(_LeastScore,"0.0")
              & "/10, " & FORMAT(_LeastBudget,"0%") & " budget adherence). "
VAR _Line5 = IF(HASONEVALUE(DimSalesTerritory[Group]),
              "" & _SelectedRegion & " - " & FORMAT(_RevShare,"0.0%")
              & " of total revenue, "
              & IF(_Variance>=0,"ahead","behind") & " budget by "
              & FORMAT(ABS(_VariancePct),"0.0%") & ", YoY "
              & IF(_YoY>=0,"▲ ", "▼ ") & FORMAT(ABS(_YoY),"0.0%") & ". ",
              "☐ Select a region for breakdown.")

RETURN

IF(ISBLANK([Total Revenue]), "☐ No regional data for selected filters.",
    _Line1 & UNICHAR(10) & _Line2 & UNICHAR(10) & _Line3
    & UNICHAR(10) & _Line4 & UNICHAR(10) & _Line5)

```

5. Executive Summary

5.1 Purpose

The **Executive Summary** is the flagship narrative measure — a full five-section strategic intelligence brief rendered in a single Smart Narrative or Card visual. All figures, region names, and category names are computed dynamically. No content is hardcoded.

5.2 Section Structure

Section	Content
☐ Financial Performance	Revenue MoM, vs budget, gross profit, margin pp shift
☐ Geographic Highlights	Top YoY region, underperforming region, consistency index
☐ Product Portfolio	Weakest margin category, standout grower, highest margin category
☐ Customer Dynamics	Active customers MoM, average revenue per customer
☐ Risks & Actions	Three prioritized action items auto-populated from worst performers

5.3 All VAR Declarations

```
VAR _Rev = [Total Revenue]
VAR _RevMoM = [Revenue MoM %]
VAR _RevYoY = [Revenue YoY %]
VAR _VarPct = [Budget Variance %]
VAR _Var$ = [Budget Variance]
VAR _GP = [Gross Profit]
VAR _GPM = [Gross Profit Margin %]
VAR _MarginPP = [Margin Change PP]
VAR _Cust = [Active Customers]
VAR _CustMoM = [Customers MoM %]
VAR _RPC = [Revenue Per Customer]
VAR _TopRegion = [Top Region by YoY Growth]
VAR _TopRegionYoY = [Top Region YoY %]
VAR _TopRegionRev = CALCULATE([Total Revenue], TOPN(1,
    ALL(DimSalesTerritory[Group]),
    CALCULATE([Revenue YoY %]),DESC))
VAR _TopRegionVarPct = CALCULATE([Budget Variance %], TOPN(1,
    ALL(DimSalesTerritory[Group]),
    CALCULATE([Revenue YoY %]),DESC))
VAR _WeakRegion = [Weakest Region by Budget Variance]
VAR _WeakRegionVar$ = [Weakest Region Budget Variance $]
VAR _WeakRegionVarPct = [Weakest Region Budget Variance %]
VAR _TopCatGrowth = [Top Category by Growth]
VAR _TopCatGrowthPct = [Top Category Growth %]
VAR _TopCatMargin = [Top Category Margin %]
VAR _WeakCatMargin = [Weakest Category by Margin]
```

```

VAR _WeakCatMarginPP = [Weakest Category Margin PP]
VAR _TopCatByMargin = [Top Category by Margin]
VAR _MostConsistent = [Most Consistent Region]
VAR _MostScore = [Most Consistent Region Score]
VAR _MostPct = [Most Consistent Region Consistency %]
VAR _LeastConsistent = [Least Consistent Region]
VAR _LeastScore = [Least Consistent Region Score]
VAR _LeastBudget = [Least Consistent Region Budget Adherence]
VAR _AvgConsistency = [Avg Regional Consistency Score]

```

5.4 Section DAX

Section 1 — Financial Performance

```

VAR _Sec1 = "□ Financial Performance" & UNICHAR(10)
& "Revenue " & FORMAT(_Rev,"$#,0.0,,") & "M – "
& IF(_RevMoM>=0,"▲ ","▼ ") & FORMAT(ABS(_RevMoM),"0.0%") & " MoM, "
& IF(_VarPct>=0,"▲ ","▼ ") & FORMAT(ABS(_VarPct),"0.0%") & " vs budget. "
& "Gross profit " & FORMAT(_GP,"$#,0.0,,") & "M at "
& FORMAT(_GPM,"0.0%") & " margin ("
& IF(_MarginPP>=0,"▲ ","▼ ") & FORMAT(ABS(_MarginPP),"0.0%") & " pp MoM)."

```

Section 2 — Geographic Highlights

```

VAR _Sec2 = "□ Geographic Highlights" & UNICHAR(10)
& _TopRegion & " leads growth ("
& IF(_TopRegionYoY>=0,"+", "") & FORMAT(_TopRegionYoY,"0.0%") & " YoY, "
& FORMAT(_TopRegionRev,"$#,0.0,,") & "M, "
& IF(_TopRegionVarPct>=0,"+", "") & FORMAT(_TopRegionVarPct,"0.0%") & " vs budget). "
& _WeakRegion & " underperformed ("
& FORMAT(_WeakRegionVarPct,"0.0%") & " vs budget, "
& FORMAT(ABS(_WeakRegionVar$),"$#,0.0,,") & "M shortfall). "
& UNICHAR(10) & "□ Consistency Index (avg "
& FORMAT(_AvgConsistency,"0.0") & "/10): "
& _MostConsistent & " most reliable (" & FORMAT(_MostScore,"0.0") & "/10, "
& FORMAT(_MostPct,"0%") & " months positive YoY). "
& _LeastConsistent & " most volatile (" & FORMAT(_LeastScore,"0.0") & "/10, "
& FORMAT(_LeastBudget,"0%") & " budget adherence) – growth sustainability at risk."

```

Section 3 — Product Portfolio

```

VAR _Sec3 = "□ Product Portfolio" & UNICHAR(10)
& _WeakCatMargin & " shows margin pressure ("
& IF(_WeakCatMarginPP>=0,"▲ ","▼ ") & FORMAT(ABS(_WeakCatMarginPP),"0.0%")
& " pp) – pricing review warranted. "
& _TopCatGrowth & " is standout grower ("
& IF(_TopCatGrowthPct>=0,"+", "") & FORMAT(_TopCatGrowthPct,"0.0%") & " YoY). "
& _TopCatByMargin & " leads on margin at " & FORMAT(_TopCatMargin,"0.0%") & ". "

```

Section 4 — Customer Dynamics

```

VAR _Sec4 = "□ Customer Dynamics" & UNICHAR(10)
& FORMAT(_Cust,"#,0") & " active customers ("
& IF(_CustMoM>=0,"▲ ","▼ ") & FORMAT(ABS(_CustMoM),"0.0%") & " MoM, "
& "avg revenue/customer " & FORMAT(_RPC,"$#,0") & ". "

```

Section 5 — Risks and Actions

```
VAR _Sec5 = "□ Risks & Actions" & UNICHAR(10)
& "1. " & _WeakRegion & " recovery plan needed ("
& FORMAT(ABS(_WeakRegionVar$), "$#,0.0,,") & "M variance). "
& "2. " & _LeastConsistent & " consistency score "
& FORMAT(_LeastScore,"0.0") & "/10 – investigate month-to-month volatility. "
& "3. " & _WeakCatMargin & " margin erosion requires pricing review. "
& "4. Accelerate " & _TopCatGrowth & " inventory investment to capture momentum."
```

RETURN Statement

```
RETURN
IF(ISBLANK(_Rev), "□ No data for selected filters.",
    _Sec1 & UNICHAR(10) & UNICHAR(10)
    & _Sec2 & UNICHAR(10) & UNICHAR(10)
    & _Sec3 & UNICHAR(10) & UNICHAR(10)
    & _Sec4 & UNICHAR(10) & UNICHAR(10)
    & _Sec5)
```

6. Strategic Health Score

6.1 Concept

The **Strategic Health Score** is a composite 0–10 index scoring the business across five equally important dimensions. Each dimension is scored 0–2 and summed to a maximum of 10. The score updates dynamically with all slicer filters.

6.2 Scoring Dimensions

Dimension	Underlying Measure	Weight
Budget Attainment	Budget Variance %	20% (2pts)
YoY Growth	Revenue YoY %	20% (2pts)
Margin Health	Gross Profit Margin % + Margin Change PP	20% (2pts)
Geographic Balance	Regional Growth Consistency Score (see Sec 7)	20% (2pts)
Customer Growth	Customers MoM %	20% (2pts)

6.3 Score Measures — Full DAX

Individual Dimension Scores

```
Measure 'Score Budget Attainment' =
SWITCH(TRUE(),
    [Budget Variance %] >= 0.10, 2,
    [Budget Variance %] >= 0, 1.5,
    [Budget Variance %] >= -0.05, 1,
    [Budget Variance %] >= -0.10, 0.5, 0)

Measure 'Score YoY Growth' =
SWITCH(TRUE(),
    [Revenue YoY %] >= 0.15, 2,
    [Revenue YoY %] >= 0.08, 1.5,
    [Revenue YoY %] >= 0, 1,
    [Revenue YoY %] >= -0.05, 0.5, 0)

Measure 'Score Margin Health' =
SWITCH(TRUE(),
    [Gross Profit Margin %] >= 0.45 && [Margin Change PP] >= 0, 2,
    [Gross Profit Margin %] >= 0.40 && [Margin Change PP] >= 0, 1.5,
    [Gross Profit Margin %] >= 0.35, 1,
    [Gross Profit Margin %] >= 0.25, 0.5, 0)

Measure 'Score Customer Growth' =
SWITCH(TRUE(),
```

```
[Customers MoM %] >= 0.10, 2,
[Customers MoM %] >= 0.05, 1.5,
[Customers MoM %] >= 0, 1,
[Customers MoM %] >= -0.05, 0.5, 0)
```

Composite Score

```
Measure 'Strategic Health Score' =
ROUND([Score Budget Attainment] + [Score YoY Growth]
    + [Score Margin Health] + [Score Geographic Balance]
    + [Score Customer Growth], 1)

Measure 'Strategic Health Score Label' =
FORMAT([Strategic Health Score], "0.0") & " / 10 - "
& SWITCH(TRUE(),
    [Strategic Health Score] >= 9, "☐ Excellent",
    [Strategic Health Score] >= 7, "☐ Strong",
    [Strategic Health Score] >= 5, "☐ Moderate",
    [Strategic Health Score] >= 3, "☐ Weak",
    "☐ Critical")
```

6.4 Score Threshold Reference

Score Range	Label & Signal
9.0 – 10.0	☐ Excellent — Peak Performance
7.0 – 8.9	☐ Strong Position
5.0 – 6.9	☐ Moderate — Monitor Closely
3.0 – 4.9	☐ Weak — Intervention Required
0.0 – 2.9	☐ Critical — Immediate Action Needed

7. Regional Growth Consistency

7.1 Concept

The Regional Growth Consistency metric scores each region not just on **how much** it grew, but **how reliably** it grew. A region spiking +30% in one month then declining in five others scores lower than one growing +8% every month. The score is a weighted composite of three sub-dimensions, each normalised to 0–1 and scaled to 0–2 for inclusion in the Health Score.

7.2 Sub-Dimension Weights

Sub-Dimension	What It Measures	Weight
Growth Rate (<code>_AvgScore</code>)	Average YoY % across all months in context — normalised against 15% ceiling	40%
Consistency (<code>_BreadthScore</code>)	Proportion of months with positive YoY growth — rewards steady growth	40%
Budget Adherence (<code>_GapScore</code>)	Gap between best and worst region consistency score — penalises inequality	20%

7.3 All Consistency Measures — Full DAX

```
Measure 'Region Months Positive YoY' =
COUNTROWS (FILTER (VALUES (DimDate[MonthNo]),
    CALCULATE ([Revenue YoY %]) > 0))

Measure 'Region Total Months' =
COUNTROWS (FILTER (VALUES (DimDate[MonthNo]),
    CALCULATE ([Total Revenue]) > 0))

Measure 'Region Consistency Rate' =
DIVIDE ([Region Months Positive YoY], [Region Total Months])

Measure 'Region Months On Budget' =
COUNTROWS (FILTER (VALUES (DimDate[MonthNo]),
    CALCULATE ([Budget Variance %]) >= 0))

Measure 'Region Budget Adherence Rate' =
DIVIDE ([Region Months On Budget], [Region Total Months])

Measure 'Region Avg YoY %' =
AVERAGEX (VALUES (DimDate[MonthNo]),
    CALCULATE ([Revenue YoY %]))

Measure 'Region Growth Score' =
```

```

MIN(DIVIDE([Region Avg YoY %], 0.15), 1)

Measure 'Region Consistency Score' = [Region Consistency Rate]

Measure 'Region Budget Score' = [Region Budget Adherence Rate]

Measure 'Regional Growth Consistency Score' =
ROUND((([Region Growth Score] * 0.40)
+ ([Region Consistency Score] * 0.40)
+ ([Region Budget Score] * 0.20), 2)

Measure 'Regional Growth Consistency Score 10' =
ROUND([Regional Growth Consistency Score] * 10, 1)

```

7.4 Fleet-Level Aggregates

```

Measure 'Most Consistent Region' =
CALCULATE(FIRSTNONBLANK(DimSalesTerritory[SalesTerritoryGroup],1),
TOPN(1,ALL(DimSalesTerritory[Group]),
CALCULATE([Regional Growth Consistency Score]),DESC))

Measure 'Most Consistent Region Score' =
CALCULATE([Regional Growth Consistency Score 10],
TOPN(1,ALL(DimSalesTerritory[Group]),
CALCULATE([Regional Growth Consistency Score]),DESC))

Measure 'Most Consistent Region Consistency %' =
CALCULATE([Region Consistency Rate],
TOPN(1,ALL(DimSalesTerritory[Group]),
CALCULATE([Regional Growth Consistency Score]),DESC))

Measure 'Least Consistent Region' =
CALCULATE(FIRSTNONBLANK(DimSalesTerritory[Group],1),
TOPN(1,ALL(DimSalesTerritory[Group]),
CALCULATE([Regional Growth Consistency Score]),ASC))

Measure 'Least Consistent Region Score' =
CALCULATE([Regional Growth Consistency Score 10],
TOPN(1,ALL(DimSalesTerritory[Group]),
CALCULATE([Regional Growth Consistency Score]),ASC))

Measure 'Least Consistent Region Budget Adherence' =
CALCULATE([Region Budget Adherence Rate],
TOPN(1,ALL(DimSalesTerritory[Group]),
CALCULATE([Regional Growth Consistency Score]),ASC))

Measure 'Avg Regional Consistency Score' =
AVERAGEX(VALUES(DimSalesTerritory[Group]),
CALCULATE([Regional Growth Consistency Score 10]))

```

7.5 Score Geographic Balance — Updated

The original **Score Geographic Balance** used *Top Region Variance Share %* as a proxy — measuring only variance concentration. The updated version uses the Regional Growth Consistency measures for a richer three-dimensional score.

```
Measure 'Regions Above Consistency Threshold' =
COUNTROWS (FILTER (VALUES (DimSalesTerritory[Group]),
    CALCULATE ([Regional Growth Consistency Score]) >= 0.6))

Measure 'Total Regions' =
COUNTROWS (VALUES (DimSalesTerritory[Group]))

Measure 'Regions Above Threshold Rate' =
DIVIDE ([Regions Above Consistency Threshold], [Total Regions])

Measure 'Consistency Score Gap' =
CALCULATE ([Regional Growth Consistency Score],
    TOPN (1, ALL (DimSalesTerritory[Group]),
        CALCULATE ([Regional Growth Consistency Score]), DESC))
- CALCULATE ([Regional Growth Consistency Score],
    TOPN (1, ALL (DimSalesTerritory[Group]),
        CALCULATE ([Regional Growth Consistency Score]), ASC))

Measure 'Score Geographic Balance' =
VAR _AvgConsistency = [Avg Regional Consistency Score]
VAR _BreadthRate = [Regions Above Threshold Rate]
VAR _Gap = [Consistency Score Gap]
VAR _AvgScore = SWITCH (TRUE (),
    _AvgConsistency >= 8, 2, _AvgConsistency >= 6, 1.5,
    _AvgConsistency >= 4, 1, _AvgConsistency >= 2, 0.5, 0)
VAR _BreadthScore = SWITCH (TRUE (),
    _BreadthRate >= 0.80, 2, _BreadthRate >= 0.60, 1.5,
    _BreadthRate >= 0.40, 1, _BreadthRate >= 0.20, 0.5, 0)
VAR _GapScore = SWITCH (TRUE (),
    _Gap <= 2, 2, _Gap <= 3, 1.5,
    _Gap <= 4, 1, _Gap <= 5, 0.5, 0)
RETURN
ROUND (( _AvgScore * 0.40) + ( _BreadthScore * 0.40) + ( _GapScore * 0.20), 2)
```

8. BenchmarkTable

8.1 Purpose

The **BenchmarkTable** is a static reference table storing industry KPI benchmarks. It eliminates hardcoded constants from DAX measures, allowing benchmarks to be updated by editing the table alone — no DAX changes required. It is also used to drive conditional formatting in card visuals (▲ Above / ▼ Below Industry Avg labels).

8.2 Table Structure

Create via Home → Enter Data in Power BI Desktop:

Metric (text)	Benchmark (decimal)
YoY Growth	0.112
Gross Margin	0.430
Budget Attainment	1.000
Customer Growth	0.050
Health Score Threshold	0.700

8.3 Lookup Measure Pattern

```
Measure 'Industry Avg YoY Benchmark' =  
CALCULATE (MAX (BenchmarkTable[Benchmark]),  
    BenchmarkTable[Metric] = "YoY Growth")  
  
Measure 'Industry Avg Gross Margin Benchmark' =  
CALCULATE (MAX (BenchmarkTable[Benchmark]),  
    BenchmarkTable[Metric] = "Gross Margin")
```

8.4 Benchmark Trend Label Measures

Used as Reference Label 2 in card visuals to show ▲ Above / ▼ Below Industry Avg:

```
Measure 'YoY vs Industry Label' =  
IF ([Revenue YoY %] >= [Industry Avg YoY Benchmark],  
    "▲ Above Industry Avg (", "▼ Below Industry Avg ("  
& FORMAT ([Industry Avg YoY Benchmark], "0.0%") & ") "  
  
Measure 'Gross Margin vs Industry Label' =  
IF ([Gross Profit Margin %] >= [Industry Avg Gross Margin Benchmark],  
    "▲ Above Industry Avg (", "▼ Below Industry Avg ("  
& FORMAT ([Industry Avg Gross Margin Benchmark], "0.0%") & ") "
```

```
Measure 'Budget Attainment vs Target Label' =
IF([Revenue Achievement %] >= 1,
    "▲ Ahead of Budget Target (", "▼ Behind Budget Target (")
& FORMAT(CALCULATE(MAX(BenchmarkTable[Benchmark]),
    BenchmarkTable[Metric]="Budget Attainment"),"0%") & ") "
```

8.5 Usage in Card Visuals

Card	Callout Measure	Reference Label (bottom)
YoY Growth	YoY Growth Label	YoY vs Industry Label
Gross Margin	Gross Profit Margin %	Gross Margin vs Industry Label
Budget Attainment	Revenue Achievement Label	Budget Attainment vs Target Label
Strategic Health Score	Health Score Label	Health Score Trend Label
Active Customers	Active Customers formatted	Customers MoM % vs target

9. Measure Dependency Map & Implementation Order

9.1 Implementation Order

Always implement in group order — each group depends on the one before it.

Group	Measures	Depends On
1 — Core Financials	Total Revenue, Budget Revenue, Gross Profit, Gross Profit Margin %	None
2 — Period Comparisons	Total Revenue PM/PY, Revenue MoM %, Revenue YoY %, Margin Change PP	Group 1 + Date Table marked
3 — Variance	Budget Variance, Budget Variance %, Total Variance All Regions, Region Variance Share %	Groups 1–2
4 — Regional Intelligence	All Top/Weakest Region measures, Region Revenue Share %	Group 3
5 — Product Intelligence	All Top/Weakest Category measures	Groups 1–2
6 — Customer	Active Customers, Active Customers PM, Customers MoM %, Revenue Per Customer	Group 2
7 — Consistency	All Region Months, Consistency Rate, Budget Adherence, Consistency Score, fleet aggregates	Groups 2–4
8 — Health Score	All Score measures, Strategic Health Score, Label	Groups 2–7
9 — BenchmarkTable	All Industry Avg lookup and trend label measures	BenchmarkTable loaded
10 — Narratives	Executive Narrative, Regional Context Narrative, Executive Summary	All above groups

9.2 Common Errors Reference

Symptom	Likely Cause	Fix
DATEADD / SAMEPERIODLASTYEAR returns blank	DimDate not marked as Date Table	Table Tools → Mark as Date Table
TOPN measure returns multiple values	Missing or ambiguous relationship	Check and activate joins in Model view
Budget measures return blank	FactInternetSalesTarget relationship inactive	Activate in Model view
Region Total Months returns incorrect values	Using ISBLANK inside CALCULATE directly	Use COUNTROWS(FILTER(..., CALCULATE([measure]) > 0))
Consistency Score Gap is negative	DESC/ASC reversed in TOPN	Swap DESC to ASC on the subtracted term

Narrative shows raw numbers not formatted	Incorrect FORMAT string pattern	Check FORMAT pattern against data magnitude
Health Score outside 0–10	Error in upstream Score measure	Validate each Score measure individually first
Emoji not rendering in visual	Font not supporting emoji	Set visual font to Segoe UI Emoji
Narrative resets to auto-generated text	Smart Narrative auto-mode re-triggered	Use Card visual or Text Box instead (Methods 3/4)

10. Quick Reference — All Measures

Core Financial

- Total Revenue
- Budget Revenue
- Gross Profit
- Gross Profit Margin %
- Total Revenue PM
- Total Revenue PY
- Revenue MoM %
- Revenue YoY %
- Gross Profit Margin % PM
- Margin Change PP
- Budget Variance
- Budget Variance %

Regional Intelligence

- Total Revenue All Regions
- Total Variance All Regions
- Region Revenue Share %
- Region Variance Share %
- Top Region by YoY Growth
- Top Region YoY %
- Weakest Region by YoY Growth
- Weakest Region YoY %
- Top Region by Budget Variance
- Top Region Budget Variance \$ / %
- Weakest Region by Budget Variance
- Weakest Region Budget Variance \$ / %
- Top Region by Variance Share
- Top Region Variance Share %
- Top Region by Margin
- Top Region Margin

Product Intelligence

- Top Category by Revenue
- Top Category Revenue Share %
- Top Category by Growth
- Top Category Growth %
- Top Category by Margin
- Top Category Margin %
- Weakest Category by Margin

- Weakest Category Margin %
- Weakest Category Margin PP

Customer

- Active Customers
- Active Customers PM
- Customers MoM %
- Revenue Per Customer

Regional Growth Consistency

- Region Months Positive YoY
- Region Total Months
- Region Consistency Rate
- Region Months On Budget
- Region Budget Adherence Rate
- Region Avg YoY %
- Region Growth Score
- Region Consistency Score
- Region Budget Score
- Regional Growth Consistency Score
- Regional Growth Consistency Score 10
- Regions Above Consistency Threshold
- Total Regions
- Regions Above Threshold Rate
- Consistency Score Gap
- Most Consistent Region
- Most Consistent Region Score
- Most Consistent Region Consistency %
- Least Consistent Region
- Least Consistent Region Score
- Least Consistent Region Budget Adherence
- Avg Regional Consistency Score

Strategic Health Score

- Score Budget Attainment
- Score YoY Growth
- Score Margin Health
- Score Geographic Balance
- Score Customer Growth
- Strategic Health Score
- Strategic Health Score Label
- Health Score Label
- Health Score Trend Label

BenchmarkTable Lookups

- Industry Avg YoY Benchmark
- Industry Avg Gross Margin Benchmark
- YoY vs Industry Label
- Gross Margin vs Industry Label
- Budget Attainment vs Target Label
- YoY Growth Label
- Revenue Achievement %
- Revenue Achievement Label
- Revenue Achievement Trend Enhanced

Context Labels

- Selected Country
- Selected Year
- Selected Month
- Filter Context Label

Narrative Measures

- Executive Narrative
- Regional Context Narrative
- Executive Summary